



Long-term multi-horizon weather prediction using geo-based physics-informed learning

Ahlam Althobaiti¹

Received: 4 June 2026 / Accepted: 4 July 2026
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Abstract

Severe atmospheric conditions may trigger extreme weather events, leading to considerable economic losses and substantial human casualties. Weather prediction supports proactive mitigation strategies, but it remains a highly challenging process due to the nonlinear interactions between the motion dynamics and thermodynamics of the atmosphere. In this work, we introduce a geo-based physics-informed predictor (GeoPIP) for long-horizon weather prediction. By integrating geo-based atmospheric representations, transformer-based temporal modelling and physics-informed regularisation, GeoPIP is designed to address long-horizon uncertainty, geographical variability and physically plausible prediction behaviour. The proposed system can perform long-term multi-horizon dual estimations of air temperature and wind speed while preserving physical consistency, delivering a transparent interpretation of the spatiotemporal estimation behaviour and the contribution of the most influential atmospheric parameters. We evaluated the performance of our system by utilising real-world measurements collected across geographically distributed weather stations in Saudi Arabia. GeoPIP yielded a low prediction error of 1.497°C for air temperature and 0.420 m/s for wind speed across large-scale meteorological environments. It has low computational complexity, and its architecture can be easily integrated into intelligent meteorological monitoring systems to realise next-generation long-horizon weather prediction.

Keywords Long-horizon weather prediction · Multi-horizon estimation · Physics-informed learning · Geo-based transformer · Air temperature prediction · Wind speed prediction

1 Introduction

Weather conditions describe the state of the atmosphere through a range of meteorological variables, including temperature, wind speed, humidity and precipitation. Severe or anomalous atmospheric conditions may trigger extreme weather events such as heatwaves, intense rainfall and typhoons (Allen et al. 2025). These events can result in considerable economic losses and substantial human casualties. For example, it has been estimated that around 489,000 heat-related deaths occur globally each year, including more than 70,000 deaths attributed to major European heatwaves (World Health Organization 2024). Therefore, weather prediction is essential for supporting proactive mitigation strategies and reducing the societal impacts of extreme weather phenomena.

Nevertheless, weather prediction, which may also be interpreted as a specialised form of video prediction (Xu et al. 2024), remains a highly challenging process due to the nonlinear interactions between the motion dynamics and thermodynamics of the atmosphere (Lin et al. 2023). Accordingly, extensive research efforts have been directed towards developing more precise prediction schemes to improve public safety and support sustainable economic development.

These efforts have generally evolved along two principal directions: numerical weather prediction and data-driven approaches. Numerical weather prediction (NWP) (Bougeault et al. 2010; Holton 1992) systems generate estimations by numerically solving the partial differential equations governing atmospheric dynamics (Nitha et al. 2025). Despite substantial advancements in recent decades, NWP systems remain constrained by high computational complexity, costly specialised infrastructure and difficulties in accurately representing composite atmospheric interactions (Ren et al. 2021). Furthermore, conventional NWP approaches often exhibit limited flexibility in uncertainty quantification and may intro-

✉ Ahlam Althobaiti
a.s.althobaiti@tu.edu.sa

¹ Department of Computer Science, College of Computers and Information Technology, Taif University, Taif 21944, Saudi Arabia

duce systematic prediction biases due to their rigid modelling structures (Brotzge et al. 2023).

By virtue of the direct relationship between machine learning technologies and data-driven analytical methodologies, the second major approach, data-driven weather prediction, has received considerable attention in numerous studies (Verma et al. 2024; Lin et al. 2026; Raissi et al. 2019; Almarashi 2023; Allen et al. 2025; Chai et al. 2021; Lin et al. 2022; Amanullah et al. 2025; Lin et al. 2023; Xu et al. 2024; Sivakrishna et al. 2025; Li et al. 2022; Wei et al. 2023). Compared with conventional NWP systems, data-driven approaches can generate rapid predictions with lower computational and resource requirements (Waqas et al. 2025). Moreover, these approaches have demonstrated competitive, and in some cases superior, estimation performance relative to NWP methods (Xu et al. 2024).

However, most existing studies remain limited to short-to-medium-range prediction horizons, predominantly focusing on single-variable predictions for the next day or, at the most, ten days ahead (Verma et al. 2024; Lin et al. 2026; Raissi et al. 2019; Almarashi 2023). We argue that such monolithic approaches are ineffective in practical prediction environments because they do not address the fundamental challenge of long-horizon prediction. Moreover, because such approaches rely purely on statistical patterns without grounding in atmospheric dynamics, their physical consistency and generalisability are fundamentally compromised (Allen et al. 2025; Chai et al. 2021; Lin et al. 2022; Amanullah et al. 2025; Lin et al. 2023; Xu et al. 2024; Sivakrishna et al. 2025; Li et al. 2022; Wei et al. 2023). Furthermore, many existing approaches do not explicitly account for geographical variability across weather station deployments, which limits their ability to adapt to region-specific atmospheric behaviours. In practice, a practical long-horizon weather prediction scheme should be capable of modelling long-range temporal dependencies, geographical atmospheric variability and physically consistent weather evolution beyond the conventional medium-range barrier.

Therefore, in this article, we aim to address these challenges and research gaps by proposing a geo-based physics-informed predictor (GeoPIP) for long-horizon weather prediction. The proposed approach incorporates geo-based atmospheric representations, transformer-based temporal modelling for capturing long-range atmospheric dependencies, and physics-informed regularisation to improve prediction reliability across extended prediction horizons. By synergising geographical context and atmospheric physical properties, GeoPIP enhances the estimation of dual weather variables (i.e., air temperature and wind speed) while maintaining physically plausible prediction behaviour. Furthermore, an attention-based interpretation mechanism makes the prediction process transparent by identifying and

analysing the contribution of influential atmospheric factors. The contributions of this article are as follows:

1. Introducing the novel GeoPIP defined by the synergy of geo-based atmospheric representation with physics-informed long-horizon weather prediction for dual weather variable estimation across station deployments.
2. Developing a physics-informed atmospheric representation strategy that incorporates geographical context, seasonal behaviour, wind circulation, dust severity and moisture-related indicators to improve the reliability and physical plausibility of long-horizon prediction.
3. Constructing an explainability mechanism to enable our system to provide a transparent interpretation of the prediction behaviour and the underlying spatiotemporal mechanism.

The remainder of this article is organised as follows. Section 2 reviews the related literature. Section 3 describes the methodology underpinning the proposed prediction system, and Section 4 outlines the evaluation methodology. Section 5 evaluates the proposed solution and demonstrates its ability to achieve low error in multi-horizon long-term weather prediction. Finally, Section 6 concludes this article.

2 Related work

In general, weather variable prediction can be categorised into two key approaches: i) pure data-driven prediction schemes (Allen et al. 2025; Chai et al. 2021; Lin et al. 2022; Amanullah et al. 2025; Lin et al. 2023; Xu et al. 2024; Sivakrishna et al. 2025; Li et al. 2022; Wei et al. 2023) and ii) hybrid physics-informed schemes (Verma et al. 2024; Lin et al. 2026; Raissi et al. 2019; Almarashi 2023). The majority of recent studies fall within the former category, focusing on predicting weather variables using purely data-driven approaches without explicitly incorporating atmospheric physical constraints. Allen et al. (2025) proposed an end-to-end deep learning prediction system based on the synergy of encoder, processor and decoder modules to estimate weather variables. Similarly, Chai et al. (2021) exploited a deep learning approach based on deformable convolutions and binocular feature extraction to profile stereoscopic visual quality perception.

Lin et al. (2022) introduced a conditional local convolution recurrent network, which integrates location-aware graph convolutions with recurrent temporal modelling to represent meteorological flow patterns and spatiotemporal dependencies across irregularly distributed weather stations. Lin et al. (2023) developed a spherical neural operator network, which formulates atmospheric dynamics through spherical partial differential equations and combines spherical and standard convolutions to learn global and local correlations.

Amanullah et al. (2025) exploited a synergy of Kalman filtering and spiking neural networks as a classification problem. Sivakrishna et al. (2025) synthesised convolutional and recurrent neural networks to capture atmospheric dependencies for regional thunderstorm estimation. Similarly, Xu et al. (2024) proposed a tensor-based convolution system to profile spatial dependencies and weather condition correlations for multiple weather predictions.

Li et al. (2022) proposed a compositional simulation platform capable of procedurally generating diverse driving scenarios to improve the generalisability of reinforcement learning systems across unseen environments. Similarly, Wei et al. (2023) proposed a recurrent network-based approach that integrates dynamic graph topology learning and spatiotemporal profiling to capture meteorological dependencies for weather estimation. However, these approaches generally rely heavily on historical data patterns and may exhibit limited robustness when profiling composite atmospheric dynamics over extended prediction horizons.

A number of studies have developed hybrid physics-informed prediction schemes attempting to integrate atmospheric physical principles into the learning process to improve the physical consistency and reliability of predictions (Verma et al. 2024; Lin et al. 2026; Raissi et al. 2019; Almarashi 2023). For example, Verma et al. (2024) considered ordinary differential equations and atmospheric advection dynamics to profile continuous atmospheric evolution and weather transport processes. Lin et al. (2026) further integrated differentiable atmospheric dynamics and physics-based parameterisation while preserving atmospheric consistency.

Raissi et al. (2019) introduced physics-informed neural networks that incorporate nonlinear partial differential equations into the training process to preserve underlying physical dynamics during prediction. Similarly, Almarashi (2023) proposed a hybrid prediction scheme based on the synergy between fuzzy logic systems and an NWP model, in which large-scale atmospheric physical dynamics are incorporated through the NWP component to improve the accuracy and reliability of daily solar radiation prediction. Nonetheless, existing physics-informed approaches primarily focus on short- to medium-range prediction scenarios, with limited consideration of geographically adaptive long-horizon multi-variable prediction.

In general, existing prediction strategies lag behind in three ways. First, a holistic approach that considers long-term multi-horizon weather prediction has not been sufficiently explored in the context of dual weather variable prediction. Second, none of the schemes proposed in previous studies can effectively adapt to the varying properties of spatiotemporal atmospheric dynamics across geographically distributed weather station deployments. Third, most existing solutions fail to incorporate physical atmospheric constraints; consideration should be given to how to preserve physically consistent prediction behaviour across extended prediction horizons.

3 GeoPIP methodology

As illustrated in Fig. 1, our proposed predictor consists of four main components: (i) feature construction, (ii) a geo-based predictor, (iii) physics-informed regulator and (iv) an

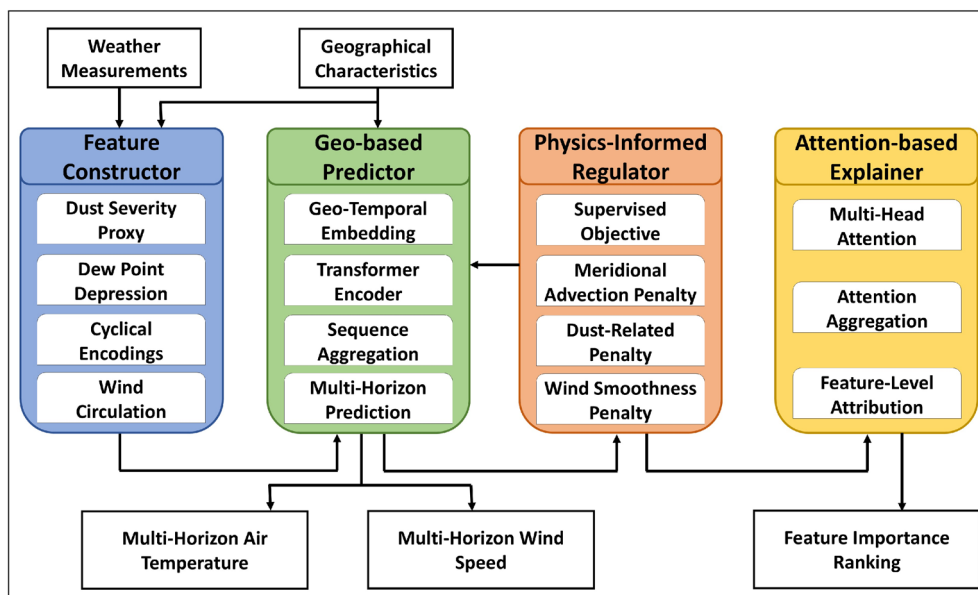
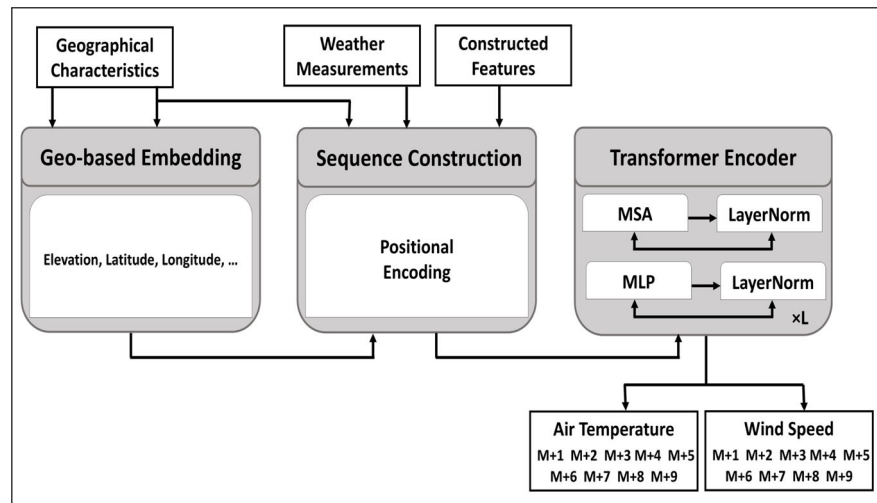


Fig. 1 Data-flow of the proposed GeoPIP

Fig. 2 Basic architecture of the geo-based predictor

attention-based explainer¹. The synergy among these components enables long-term prediction within a closed-loop system, in which spatial context informs temporal processing, physical constraints regularise the learning process, and attention attribution provides a diagnostic layer for meteorological validation.

3.1 Feature construction

To capture the spatiotemporal dynamics underlying physical atmospheric interactions, GeoPIP employs a feature construction process that transforms raw weather measurements from geographically distributed meteorological stations into an augmented representation. Specifically, to improve the model's physical expressiveness, GeoPIP incorporates meteorologically motivated proxies associated with dust events. This augmented representation is formulated as:

$$\mathcal{I}_{\text{dust}} = \frac{W}{V_{\text{vis}} + 1} \quad (1)$$

where W denotes wind speed and V_{vis} represents visibility distance. This proxy is motivated by the established relationship between strong surface winds, airborne dust concentration and visibility degradation during dust events (Assiri et al. 2025).

Similarly, atmospheric moisture saturation conditions are represented using a dew point depression formulation:

$$\Delta T_{\text{dp}} = T_{\text{air}} - T_{\text{dew}} \quad (2)$$

where T_{air} and T_{dew} denote air temperature and dew point temperature, respectively. This representation is physically

associated with atmospheric dryness, humidity variability and aerosol activity in arid and semi-arid environments (Assiri et al. 2025; Wallace and Hobbs 2006).

Furthermore, GeoPIP incorporates cyclical temporal encodings based on sinusoidal sine and cosine representations, alongside wind-vector representations, to preserve seasonal continuity and large-scale atmospheric circulation characteristics. The proposed feature construction process generates a high-dimensional spatiotemporal representation capable of modelling nonlinear atmospheric interactions across extended prediction horizons.

3.2 Geo-based predictor

Due to its ability to simultaneously model long-range temporal dependencies and geographically conditioned atmospheric interactions, this module adopts a geo-based transformer architecture to predict dual weather variables (i.e. air temperature and wind speed). The proposed architecture offers several advantages over conventional prediction models, which generally exhibit limited adaptability to geographically distributed atmospheric variability and composite spatiotemporal dependencies across extended estimation horizons (Dai et al. 2021). Hence, the proposed geo-based predictor learns geographically aware representations that preserve long-horizon atmospheric consistency.

The architecture of the proposed geo-based predictor is illustrated in Fig. 2. The model utilises the augmented spatiotemporal representation described in Section 3.1 as the dynamic input, while incorporating static geographical information as a high-level semantic prior. To embed the augmented representation within the latent transformer space, a temporal projection function ϕ_{time} is defined as:

$$\phi_{\text{time}} : \mathbb{R}^F \rightarrow \mathbb{R}^{d_{\text{model}}} \quad (3)$$

¹ The complete prediction system is available on <https://github.com/Ahham-Althobaiti/GeoPIP>.

where F denotes the dimensionality of the augmented feature space, and d_{model} represents the latent embedding dimension utilised by the transformer encoder.

The transformer encoder receives a geographically conditioned sequence representation constructed from the projected temporal embeddings, the geo-based embedding and a global representation, as follows:

$$Z_0 = [z_{\text{cls}} \parallel z_{\text{geo}} \parallel \phi_{\text{time}}(x_1) \parallel \cdots \parallel \phi_{\text{time}}(x_T)] + E_{\text{pos}} \quad (4)$$

where $z_{\text{cls}} \in \mathbb{R}^{d_{\text{model}}}$ is a global representation embedding employed for sequence-level aggregation, $z_{\text{geo}} = \phi_{\text{geo}}(g)$ denotes the geo-based embedding, and E_{pos} represents the positional encoding matrix. In this work, the geographical input vector g is defined as:

$$g = [e, \sin(\phi), \cos(\phi), \sin(\ell), \cos(\ell)] \quad (5)$$

where e denotes station elevation, ϕ denotes latitude, and ℓ denotes longitude. No climate-zone label or additional external geographical classification was used in the geographical embedding.

The geographically conditioned latent representations are recursively refined through stacked transformer encoder layers, where the latent representations are recursively updated as follows:

$$Z'_l = \text{LayerNorm}(Z_{l-1} + \text{MSA}(Z_{l-1})) \quad (6)$$

$$Z_l = \text{LayerNorm}(Z'_l + \text{FFN}(Z'_l)) \quad (7)$$

where LayerNorm is residual normalisation and MSA($\cdot$) and FFN($\cdot$) are the multi-head self-attention and position-wise feed-forward operations, respectively.

The final latent representation is subsequently mapped into the dual weather prediction space corresponding to air temperature and wind speed through a multi-layer perceptron (MLP), formulated as follows:

$$\hat{Y} = \text{MLP}(z_{\text{cls}}^{(L)}) \quad (8)$$

where $z_{\text{cls}}^{(L)}$ denotes the final global latent representation obtained from layer L .

3.3 Physics-informed regulator

The objective of this module is to regularise the optimisation process using physically motivated constraints, thereby encouraging the proposed GeoPIP to preserve physically consistent prediction behaviour across extended estimation horizons.

In particular, the primary optimisation objective of the proposed prediction system is defined using the Smooth

Huber function computed across the multi-horizon estimation sequence with an exponential horizon weighting scheme, as follows:

$$\mathcal{L}_{\text{sup}} = \frac{1}{H \cdot D} \sum_{h=1}^H w_h \text{SmoothL1}(\hat{y}_h, y_h) \quad (9)$$

where H and D are the prediction horizon and the dimensionality of the target variables, respectively, and $\hat{y}_h$ and y_h are the predicted and ground-truth states at prediction step h . The term w_h is a horizon-dependent weighting factor.

To enforce physical consistency, a meridional advection penalty is defined as:

$$\mathcal{L}_{\text{adv}} = \frac{1}{H-1} \sum_{h=2}^H \gamma^h \Omega(g) M_{\text{season}}(h) \left[\max(0, \Delta \hat{T}_h) \sigma\left(\frac{V - v_{\tau}}{s_v}\right) + \max(0, -\Delta \hat{T}_h) \sigma\left(\frac{-V - v_{\tau}}{s_v}\right) \right] \quad (10)$$

where V , v_{τ} and $\Delta \hat{T}_h$ are the meridional wind component, the critical threshold and the predicted temperature variation, respectively, and γ^h , $\Omega(g)$, $M_{\text{season}}(h)$ and $\sigma(\cdot)$ are the temporal decay, spatial weighting, seasonal masking and sigmoid functions, respectively.

This formulation is physically motivated by the well-established relationship between large-scale wind advection and temperature variability, where directional wind flow is closely associated with temperature changes and extremes. Such interactions have been widely reported across different climatic regions, including recent studies on temperature variability and extremes (Almazroui 2020; Aldughairi 2025). The constraint enforces directional consistency between meridional wind flow and temperature changes while accounting for spatial and temporal variability.

Similarly, a dust-related constraint penalty is defined as:

$$\mathcal{L}_{\text{dust}} = \frac{1}{H-1} \sum_{h=2}^H \gamma^h \Omega_{\text{dust}}(g) M_{\text{season}}(h) \times \max\left(0, |\Delta \hat{T}_h| - \tau_{\text{elev}}(g)\right) \times \sigma\left(\frac{\mathcal{I}_{\text{dust}} - \delta_{\text{dust}}}{s_d}\right) \quad (11)$$

where $\mathcal{I}_{\text{dust}}$, δ_{dust} and $\tau_{\text{elev}}(g)$ are the dust severity proxy, the critical threshold and the elevation-dependent thermal tolerance parameter, respectively, and γ^h , $\Omega_{\text{dust}}(g)$, $M_{\text{season}}(h)$ and $\sigma(\cdot)$ are the temporal decay, spatial weighting, seasonal masking and sigmoid functions, respectively.

This constraint is informed by the established physical characteristics of dust events, where strong surface winds, reduced visibility and abrupt temperature variations are often coupled across arid and semi-arid regions (Assiri et al.

2025). The penalty limits unrealistic temperature fluctuations under dusty conditions and ensures consistency with observed atmospheric behaviour.

In addition, a wind smoothness penalty is introduced to discourage unrealistic temporal discontinuities in predicted wind speed, as follows:

$$\mathcal{L}_{\text{wind}} = \frac{1}{H-1} \sum_{h=2}^H \gamma^h \max(0, |\Delta \hat{W}_h| - \tau_w) \quad (12)$$

where $\Delta \hat{W}_h = \hat{W}_h - \hat{W}_{h-1}$ denotes the predicted wind increment and τ_w is a predefined threshold controlling allowable wind variability.

The final composite regulation function is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{sup}} + \lambda (\omega_{\text{adv}} \mathcal{L}_{\text{adv}} + \omega_{\text{dust}} \mathcal{L}_{\text{dust}} + \omega_{\text{wind}} \mathcal{L}_{\text{wind}}) \quad (13)$$

where λ , ω_{adv} , ω_{dust} and ω_{wind} are hyperparameters controlling the contribution of the physics-based regularisation components.

The complete end-to-end optimisation process for the geo-based predictor is summarised in Algorithm 1. As illustrated in this algorithm, the training procedure iteratively optimises the proposed GeoPIP by combining supervised prediction objectives with physics-informed atmospheric constraints. During each training iteration, the model generates multi-horizon predictions, extracts the required physical and geographical variables for constraint computation, evaluates the total regulation function and updates the network parameters through a backpropagation operation.

Algorithm 1 Training Procedure.

Require: Training dataset $\mathcal{D}$

Require: Physics regularisation coefficients λ , ω_{adv} , ω_{dust} , ω_{wind}

Ensure: Optimised GeoPIP parameters Θ

- 1: Initialise the GeoPIP parameters Θ
 - 2: **for** each training epoch **do**
 - 3: **for** each mini-batch (X_b, y_b) **do**
 - 4: Generate multi-horizon weather predictions $\hat{y}_b \leftarrow f_{\Theta}(X_b)$
 - 5: Recover the atmospheric variables required for physics constraint evaluation
 - 6: Compute the temporal variations $\Delta \hat{T}_h$ and $\Delta \hat{W}_h$
 - 7: Compute the supervised objective $\mathcal{L}_{\text{sup}}$
 - 8: Evaluate the advection and dust constraints using (10) and (11)
 - 9: Evaluate the wind smoothness constraint using (12)
 - 10: Construct the composite physics-informed objective using (13)
 - 11: Update the GeoPIP parameters through gradient-based optimisation
 - 12: **end for**
 - 13: **end for**
 - 14: **return** Θ
-

3.4 Attention-based explainer

This component of the proposed GeoPIP is designed to provide the interpretability necessary to bridge the gap between predictive performance and scientific justification. To this end, multi-head attention attribution is employed to audit the model's internal decision-making processes, thereby ensuring that physical constraints are reflected coherently in its internal representational focus.

Let $A_h^{(l)} \in \mathbb{R}^{(T+2) \times (T+2)}$ denote the attention matrix of head h at layer l . The input sequence comprises a geo-based embedding, a global representation embedding and temporal embeddings. The attention weights are first consolidated by averaging across all layers and heads as follows:

$$A_{\text{avg}} = \frac{1}{L \cdot H} \sum_{l=1}^L \sum_{h=1}^H A_h^{(l)} \in \mathbb{R}^{(T+2) \times (T+2)} \quad (14)$$

where L denotes the total number of transformer layers, H is the number of attention heads per layer and $A_h^{(l)}$ denotes the attention weight matrix of the h -th head at the l -th layer.

A global importance score is then derived by extracting the attention allocated from the global representation embedding towards each position in the sequence. This quantifies the proportion of the attention budget directed to the geo-based embedding relative to the temporal embeddings, as follows:

$$\alpha_j = A_{\text{avg}}[z_{\text{cls}}, j], \quad \hat{\alpha}_j = \frac{\alpha_j}{\sum_k \alpha_k} \quad (15)$$

where α_j denotes the raw attention score from the z_{cls} embedding to position j , and k indexes all positions in the sequence $\{z_{\text{geo}}, \text{Time}_1, \dots, \text{Time}_T\}$.

To further decompose this global signal into feature-level attributions, the normalised temporal attention scores are employed to weight the absolute mean magnitude of each input variable across the batch, yielding a feature importance vector as:

$$I_f = \sum_{t=1}^T \hat{\alpha}_t \cdot \left(\frac{1}{B} \sum_{b=1}^B |X_{b,t,f}| \right), \quad \forall f \in \{1, \dots, F\} \quad (16)$$

where $X_{b,t,f} \in \mathbb{R}$ is the value of feature f at time step t for the b -th sample in the batch. The inner term $\frac{1}{B} \sum_{b=1}^B |X_{b,t,f}|$ is the mean absolute magnitude of feature f at time step t .

Finally, the resulting feature attribution vector is normalised to the unit interval as follows:

$$\tilde{I}_f = \frac{I_f}{\max_{f'} I_{f'}} \quad (17)$$

where B denotes the batch size, T denotes the input sequence length, F denotes the number of input features and $\tilde{I}_f \in$

$[0, 1]$ denotes the final normalised importance score assigned to feature f ; $\tilde{I}_f = 1$ indicates the most influential feature, while values approaching zero indicate negligible contribution to the model's predictive focus.

The complete attribution mechanism is formalised in Algorithm 2. As illustrated in this algorithm, the proposed explainer first extracts and aggregates the multi-head attention representations across all transformer layers to construct a unified attention map. Subsequently, a temporal attention distribution is derived from the global representation embedding z_{cls} . The resulting attention distribution is then integrated with the meteorological feature representations to compute feature-level attribution scores across the input sequence. Finally, the attribution scores are globally normalised to generate an interpretable importance profile that highlights the dominant atmospheric predictors governing the prediction behaviour of GeoPIP.

Algorithm 2 Attention-Based Attribution.

Require: Trained GeoPIP model f_Θ and input sequence $X \in \mathbb{R}^{B \times T \times F}$

Ensure: Feature attribution vector $I \in \mathbb{R}^F$

- 1: Extract multi-head attention representations $A_h^{(l)}$
- 2: Compute the averaged attention representation A_{avg} using (14)
- 3: Derive the normalised temporal attention scores $\hat{\alpha}$ using (15)
- 4: Compute the mean absolute feature representation:

$$X_{abs} = \frac{1}{B} \sum_{b=1}^B |X_b|$$

- 5: Compute the feature attribution scores I_f using (16)
 - 6: Normalise the attribution vector using (17)
 - 7: **return** I
-

4 Evaluation methodology

To validate our work, we used climate datasets collected from different stations across Saudi Arabia, provided by the King Abdullah Petroleum Studies and Research Center (KAPSARC), a leading research institution specialising in energy economics, sustainability and data-driven policy analysis². The datasets include hourly observations of meteorological factors such as wind, sky conditions, visibility, air temperature, dew point and sea level pressure, covering the most recent five years, with additional historical data available. In this work, we aggregated the measurements into monthly intervals and processed them using a sliding window approach with a look-back period of 12 months and a prediction horizon of 9 months to capture seasonal

patterns. In addition to meteorological measurements, the dataset includes information regarding station geolocation (e.g. longitude and latitude).

To demonstrate the effectiveness of the proposed system, we conducted a performance comparison across five prediction baselines grouped into three categories: (i) conventional climatology as a statistical reference; (ii) machine learning models, represented by extreme gradient boosting (XGBoost) and support vector machine (SVM); and (iii) deep learning models, represented by MLP and long short-term memory (LSTM). This evaluation excludes statistical models, including persistence, seasonal autoregressive integrated moving average and exponential smoothing, as such methods rely on long, continuous time series with stable and well-defined patterns. Our dataset is characterised by multiple stations, short time series per station and sophisticated multivariate dynamics, which often leads to poor convergence and unreliable performance for these models.

For this comparison, we utilise the following performance metrics.

1. The RMSE is defined as:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (18)$$

2. The MAE is defined as:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (19)$$

3. The coefficient of determination (R^2) is defined as:

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (20)$$

4. Willmott's Index (WI) is defined as:

$$\text{WI} = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (|\hat{y}_i - \bar{y}| + |y_i - \bar{y}|)^2} \quad (21)$$

5. The Absolute Percentage Bias (AbsPBIAS) is defined as:

$$\text{AbsPBIAS} = 100 \times \left| \frac{\sum_{i=1}^N (\hat{y}_i - y_i)}{\sum_{i=1}^N y_i} \right| \quad (22)$$

6. The Explained Variance Score (EVS) is defined as:

$$\text{EVS} = 1 - \frac{\text{Var}(y_i - \hat{y}_i)}{\text{Var}(y_i)} \quad (23)$$

² Explore: <https://datasource.kapsarc.org/explore/assets/saudi-hourly-weather-data/>

where y_i represents the true value, $\hat{y}_i$ represents the predicted value, $\bar{y}$ is the mean of the true values, N is the total number of samples, and $\text{Var}(\cdot)$ denotes the variance operator. Here, RMSE and MAE are expressed in the units of the predicted variable ($^{\circ}\text{C}$ for air temperature and m/s for wind speed), while R^2 , WI and EVS are dimensionless, and AbsPBIAS is reported as a percentage (%).

7. **Computational complexity:** To measure the inference time required to obtain predictions on the test data in milliseconds.

It is worth mentioning that the computational complexity excludes the grid-search process utilised to train and fine-tune hyperparameters. It transforms a hyperparameter domain into a grid and then traverses each point on the grid to obtain the optimal classifier parameters. This strategy is straightforward, and the optimal search speed is reached reasonably quickly (Althobaiti et al. 2023). In addition, the optimal hyperparameters are determined independently, enabling simultaneous optimisation (Althobaiti et al. 2021). Table 1 illustrates the results of the grid-search process for each prediction algorithm, including the number of models required for multi-horizon and dual-target prediction.

As illustrated in Table 1, GeoPIP employs a single unified model to simultaneously predict all target variables across all prediction horizons. In contrast, benchmark methods such as SVM and XGBoost require multiple output-wise estimators (18 models for two target variables across nine prediction horizons), whereas MLP and LSTM utilise a single multi-output model. The geographical and seasonal configurations of GeoPIP were motivated by the regional climatological characteristics of Saudi Arabia, which exhibit pronounced spatial and seasonal variability across meteorological stations (Almazroui 2020; Aldughairi 2025; Wallace and Hobbs 2006), whereas the final values of the physics-informed

parameters were determined through empirical sensitivity analyses and validation-based optimisation.

As part of our evaluation, we analysed the sensitivity of the proposed GeoPIP at two levels. The first, system-level sensitivity, was examined to reflect its robustness and to quantify the individual contribution of each of the main components of GeoPIP. To this end, we introduced three component-level ablation cases. In the first case, the physics-informed regulation was removed, and the model was trained employing only the standard supervised penalty to assess the effect of the physical regularisation terms. In the second case, the feature construction module was removed so the model was trained utilising only the basic input variables without the engineered temporal, seasonal and physically motivated features. In the third case, the geo-based embedding was removed to evaluate the importance of spatial conditioning within the transformer architecture. Collectively, these ablation cases allowed us to determine whether the final prediction performance was driven by the complete integration of physical constraints, constructed spatiotemporal predictors and geographic embedding rather than by any single isolated component.

The second level of sensitivity considered the attention-based explainer and proceeded through feature-level sensitivity analyses. Specifically, a horizon-wise feature-occlusion analysis was conducted, in which the most influential features were identified and masked independently at each prediction step, from step 1 to step 9 (i.e. month 1 to month 9), and for each prediction target. The resulting changes in predictive performance at each horizon were then compared against a baseline obtained by randomly masking the same number of features. This sensitivity validation ensured that the highlighted features were genuinely grounded in the underlying system logic at each prediction step rather than being artefacts of stochastic attention weights.

Table 1 Hyper-parameters

Algorithm	Number of Models	Hyper-parameters
GeoPIP	Dual	Transformer encoder layers=9, Attention heads=8, Model dimension d_{model} =256, Batch size=32, Optimizer=AdamW, Learning rate= 1×10^{-4} , Number of epochs=150, $\lambda = 0.5$, $\omega_{adv} = 0.30$, $\omega_{dust} = 0.20$, $\omega_{wind} = 0.40$, $v_r = 2.0$, $s_v = 0.5$, $\delta_{dust} = 0.5$, $s_d = 0.2$, $\tau_w = 3.0$, $\Omega(g) = \text{clip}(1 + 0.35 \sigma(\frac{e-600}{400}), 0.75, 1.35)$, $\Omega_{dust}(g) = \text{clip}(2 - \Omega(g), 0.7, 1.3)$, $\tau_{elev}(g) = 5.0 + 0.002 \text{clip}(e, 0, 2000)$
SVM	18	Kernel=RBF, Regularisation parameter $C = 10$
MLP	Dual	Hidden layers=(128, 64,8), Activation=ReLU, Solver=Adam, Learning rate= 1×10^{-6}
XGBoost	18	Number of estimators=100, Maximum depth=50
LSTM	Dual	Hidden size=128, Number of LSTM hidden units=64, Batch size=32, Optimizer=Adam, Learning rate= 1×10^{-4} , Number of epochs=150

Table 2 Performance of GeoPIP and different prediction algorithms

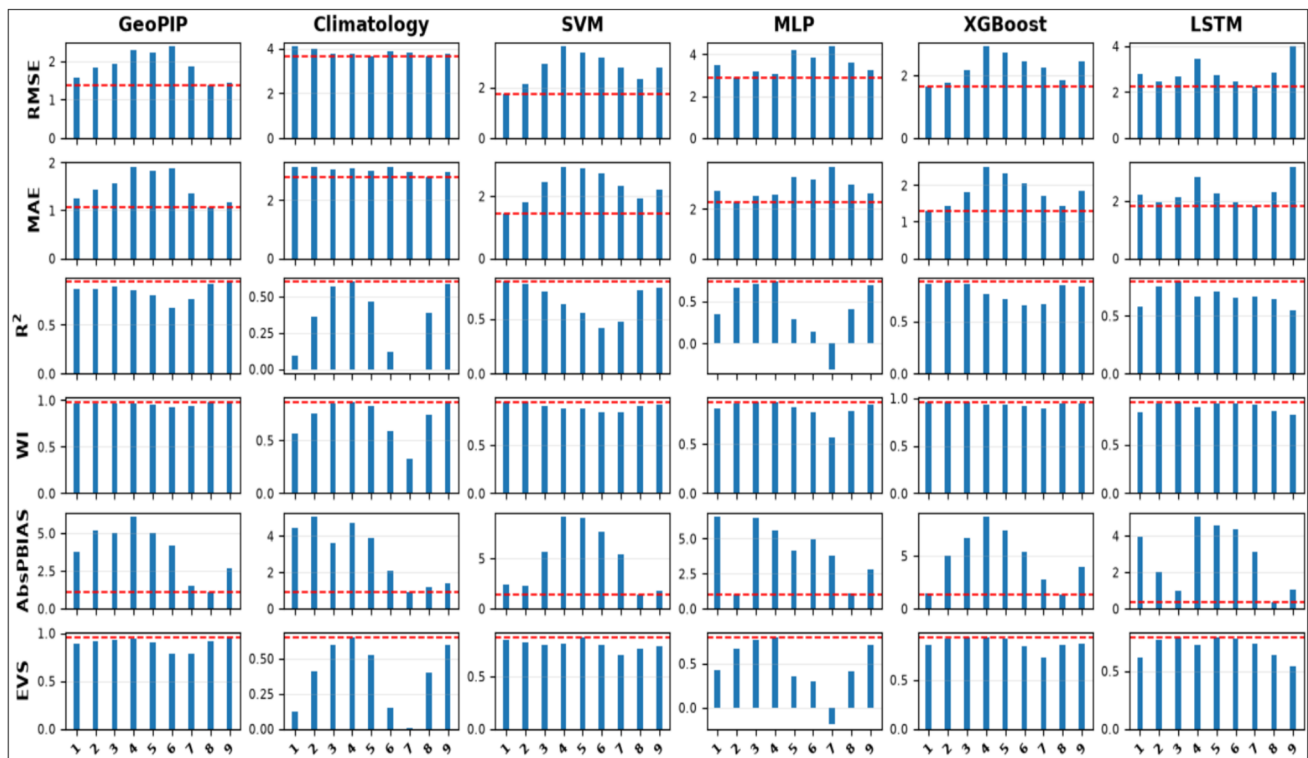
Algorithm	Air Temperature						Wind Speed					
	RMSE	MAE	R^2	WI	AbsPBIAS	EVS	RMSE	MAE	R^2	WI	AbsPBIAS	EVS
GeoPIP	1.909	1.497	0.929	0.982	2.782	0.941	0.537	0.420	0.488	0.866	3.034	0.511
Climatology	3.8322	3.0368	0.7148	0.9155	2.2033	0.7221	0.7092	0.5583	0.1073	0.4339	2.3735	0.1211
SVM	2.8138	2.2964	0.8462	0.9521	4.4281	0.8758	0.5475	0.4137	0.4679	0.7692	4.5863	0.5197
MLP	3.5843	2.8884	0.7505	0.9365	3.0458	0.7645	0.5968	0.4688	0.3678	0.8092	3.5021	0.3980
XGBoost	2.2856	1.8261	0.8985	0.9721	3.5197	0.9172	0.6641	0.5262	0.2171	0.5393	3.2426	0.2430
LSTM	2.8851	2.3018	0.8383	0.9503	1.5812	0.8421	0.5497	0.4208	0.4636	0.8084	2.6942	0.4815

5 Results

Following the evaluation methodology in Section 4, the GeoPIP formulation was evaluated against the conventional climatology, SVM, MLP, XGBoost, and LSTM benchmarks. As summarised in Table 2, GeoPIP consistently outperformed these benchmarks in air temperature prediction, attaining the lowest RMSE of 1.909 and MAE of 1.497, together with the highest R^2 of 0.929, WI of 0.982, and EVS of 0.941. Although XGBoost achieved the second-highest R^2 score of 0.8985 for air temperature prediction, its corresponding RMSE of 2.2856 and MAE of 1.8261 remained considerably higher than those of GeoPIP, underscoring GeoPIP's superior overall predictive accuracy. The only exception in the air temperature results was AbsPBIAS,

for which LSTM attained a marginally lower value of 1.5812 than GeoPIP (2.782%), indicating slightly lower systematic prediction bias.

A broadly similar pattern emerged for wind speed prediction, where GeoPIP achieved the lowest RMSE of 0.537, together with the highest R^2 value of 0.488 and WI value of 0.866 of all the models. The SVM and LSTM formulations also achieved relatively competitive results for wind speed prediction. In particular, SVM attained a marginally lower MAE of 0.4137 and a higher EVS of 0.5197 than GeoPIP, which recorded corresponding values of 0.420 and 0.511, respectively. Here, the poor performance of XGBoost reflects the limited ability of tree-based ensemble methods to capture the sequential temporal dependencies inherent in multi-horizon wind estimation, particularly for non-stationary meteorologi-

**Fig. 3** Horizon-wise performance for the air temperature prediction

cal time series without an explicit temporal structure (Ramraj et al. 2016). Nevertheless, GeoPIP's consistently mostly superior RMSE, R^2 , and WI values indicate stronger overall predictive accuracy and reliability for wind speed estimation compared with the individual benchmark models.

The superior performance of the GeoPIP formulation can be attributed to its use of multi-head self-attention, which enables the model to capture important temporal information from climate-related input variables and learn long-range dependencies across the input sequence (Ghimire et al. 2023). Unlike conventional models that primarily rely on local or fixed temporal relationships, the attention mechanism allows GeoPIP to assign varying levels of importance to relevant historical observations and constructed meteorological features. This enables the model to better represent delayed and cumulative weather effects across multiple prediction horizons, thereby improving predictive accuracy and robustness in air temperature and wind speed estimation.

GeoPIP maintained stable long-horizon behaviour, particularly for air temperature prediction, as depicted in Figs. 3 and 4. Across the prediction horizons, GeoPIP maintained air temperature RMSE values between approximately 1.389 and 2.327, while preserving competitive R^2 values ranging from 0.704 to 0.966 and a WI of 0.982 and an EVS of 0.941. This superior performance can be attributed to the feature construction module discussed in Section 3, which effectively enhances the multi-horizon meteorological prediction.

This module constructs a set of variables that characterise temporal, seasonal, and geographical weather behaviours to improve the prediction of air temperature and wind speed variables. The dataset included four main station-level and meteorological characteristics: geographical location, elevation, seasonal indicators, and weather-derived variables. For air temperature prediction, the variables derived from the temporal and geographical features were instrumental in profiling normal meteorological behaviour associated with temperature variation. Therefore, the objective of constructing informative features for air temperature prediction was achieved, leading to improved prediction performance.

In this scenario, it is necessary to employ additional physically meaningful features in the procedure to improve the prediction of wind speed measurements, especially over the middle prediction horizons. This observation is consistent with previous nonlinear and non-stationary time-series prediction studies, where decomposition and optimisation-based learning have been used to improve robustness under complex temporal dynamics (Zhang et al. 2020; Hong et al. 2019). However, a trade-off must be made between the efficiency benefit and the issue of overfitting, and the addition of features should lead to a prediction strategy that is specifically tailored to suit particular data conditions and settings, which limits its generalisability (Althobaiti et al. 2023). Overall, these findings indicate that our feature construction module plays a significant role in improving prediction performance,

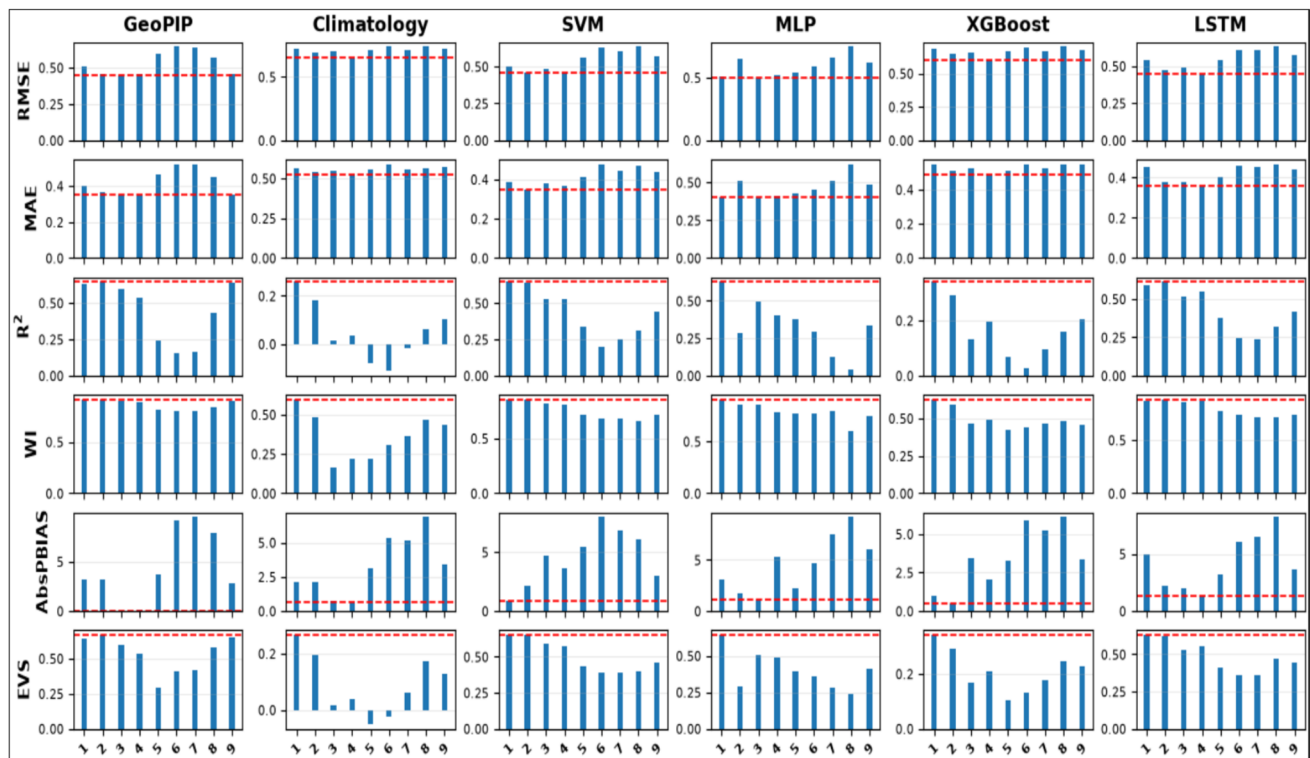
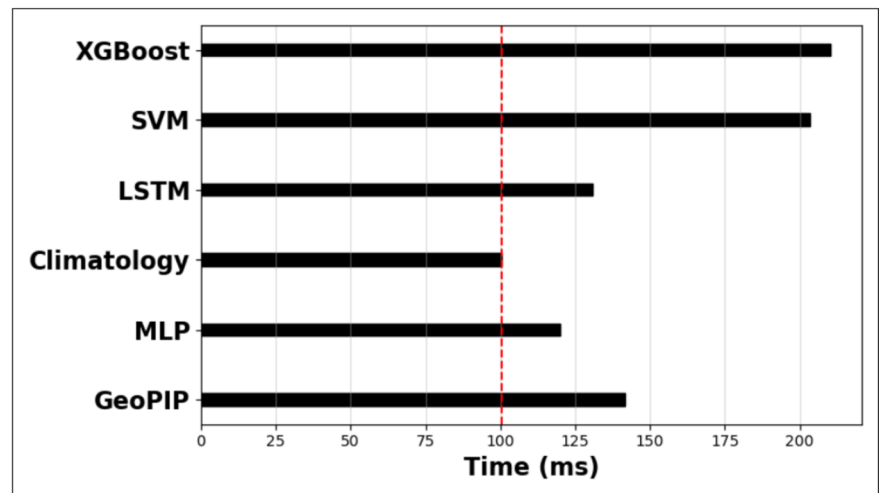


Fig. 4 Horizon-wise performance for the wind speed prediction

Fig. 5 Computational complexity time of different prediction algorithms



but it should be trained with meteorological and geographical characteristics that reflect behavioural similarities in temperature and wind speed patterns.

Apart from its high prediction accuracy, the proposed GeoPIP formulation operates with a relatively competitive computational time compared with other schemes³, as depicted in Fig. 5. This is because SVM and XGBoost models require independently optimised learning structures and lack the unified geo-based prediction mechanism employed in the proposed system, meaning they incur additional computational overheads. As demonstrated in Table 1, the formulation required 18 independently trained models to handle the multiple targets and prediction horizons, whereas the proposed GeoPIP system employed a unified dual-output architecture to simultaneously predict both air temperature and wind speed measurements across all prediction horizons within a single training procedure. In addition, compared with conventional deep learning approaches such as LSTM, GeoPIP avoids repeatedly calculating redundant spatiotemporal representations through shared learning layers, which significantly improves the efficiency of simultaneous prediction tasks. This demonstrates the efficacy of employing a geo-based deep learning technique instead of conventional prediction techniques in our system, as we needed to train only one model with multiple outputs to address both air temperature and wind speed prediction tasks simultaneously across multiple prediction horizons.

For the attention-based explainer, at the feature level, Table 3 reveals distinct spatiotemporal predictor behaviours for air temperature and wind speed prediction across different prediction horizons. Air temperature prediction is primarily governed by historical temperature statistics, seasonal

harmonic embeddings, terrain elevation and wind-vector representations. The repeated appearance of zonal meridional wind-vector components demonstrates the importance of physically meaningful circulation representations within the predictive behaviour. In contrast, wind speed prediction exhibits a more heterogeneous and locally driven predictor structure, where dust severity indicators, visibility distance, dew point depression, meridional wind-vector components and, to a lesser extent, zonal components emerge as recurring explanatory variables across multiple horizons. The persistent contribution of dust severity and dew point depression features provides further evidence for the effectiveness of the proposed physics-informed feature construction strategy, described in Section 3, in capturing the aerosol interactions and atmospheric dryness effects in arid environments.

These findings are consistent with reported seasonal temperature variability studies, which highlight warming trends and the role of elevation and regional circulation in shaping temperature extremes (Almazroui 2020). They are also consistent with dust-climatology studies, where dust events are associated with strong winds, visibility reduction, humidity variation, pressure changes, temperature drops and aerosol transport in arid and semi-arid regions (Assiri et al. 2025). Overall, these findings confirm that the proposed attention-based explainer enables GeoPIP to produce explanations consistent with established climatological studies, thereby providing an interpretable, horizon-adaptive and physics-consistent prediction mechanism that extends beyond conventional fixed-input prediction architectures.

The component-level sensitivity analysis demonstrates that the complete GeoPIP configuration provides the most stable and balanced overall prediction performance across both meteorological variables, as shown in Table 4. For air temperature prediction, GeoPIP records an RMSE of 1.909, an MAE of 1.497 and an R^2 of 0.929, together with a WI of 0.982, an AbsPBIAS of 2.782 and an EVS of 0.941. Although

³ On a64-bit Windows operating system with an Intel Core i7 CPU with a 2.80 GHz clock cycle and 32 GB of RAM.

Table 3 Attention-based top feature frequency analysis

Feature	Air Temperature	Wind Speed
Temperature Mean (6-Month Rolling)	7	0
Zonal Wind Component (3-Month Rolling)	7	2
Monthly Seasonal Sine Component	4	0
Monthly Seasonal Cosine Component	4	5
Terrain Elevation	5	0
Temperature Mean (5-Month Lag)	4	0
Meridional Wind Component (3-Month Rolling)	4	4
Dust Severity Index (3-Month Rolling)	2	5
Wind Speed Mean (3-Month Rolling)	1	6
Temperature Mean (7-Month Lag)	0	5
Visibility Distance Mean	0	4
Dew Point Depression (1-Month Lag)	0	3

removing the geo-based embedding produces a slightly lower temperature RMSE, GeoPIP achieves stronger overall reliability by maintaining the highest R^2 , WI and EVS values, as well as the lowest AbsPBIAS. For wind speed prediction, GeoPIP achieves an RMSE of 0.537, an MAE of 0.420, an R^2 of 0.488, a WI of 0.866, an AbsPBIAS of 3.034 and an EVS of 0.511. Removing the physics-informed regulation increases the temperature RMSE from 1.909 to 2.188 and reduces the corresponding R^2 from 0.929 to 0.792, while also reducing the WI from 0.982 to 0.949 and the EVS from 0.941 to 0.877. Although the wind-speed RMSE and MAE are slightly lower without the physics-informed regulation, GeoPIP retains higher R^2 and WI values and a lower AbsPBIAS, indicating more reliable and less biased predictive behaviour. This observation is consistent with the well-established benefit of physically constrained learning for atmospheric prediction generalisation and stability (Verma et al. 2024).

The largest degradation was observed when the feature construction module was removed; the temperature RMSE increased sharply to 6.768, while the corresponding R^2 , WI and EVS decreased to -0.902 , 0.570 and -0.515 , respectively. Similarly, the wind speed RMSE increased substantially to 0.877, while the corresponding R^2 , WI and EVS decreased to -0.457 , 0.477 and -0.372 , respectively. These findings confirm that the engineered temporal, seasonal and physically motivated predictors are fundamental

for representing the underlying spatiotemporal atmospheric dynamics within GeoPIP. Removing the geo-based embedding produced a slightly lower RMSE for air temperature, but it reduced the reliability indicators, including R^2 , WI and EVS, and increased the AbsPBIAS from 2.782 to 3.767. For wind speed prediction, removing the geo-based embedding also reduced the R^2 from 0.488 to 0.455 and the WI from 0.866 to 0.855, while increasing the AbsPBIAS from 3.034 to 4.697. This demonstrates that spatial conditioning contributes meaningful geographic context to the GeoPIP architecture and improves the stability and bias control of regional atmospheric prediction. Spatially structured atmospheric representations have also been identified as critical for prediction stability in large-scale data-driven systems (Allen et al. 2025). Overall, the sensitivity analysis confirms that GeoPIP's prediction capability is due to its synergistic integration of feature construction, physics-informed learning and geo-based embedding rather than any individual component in isolation.

The results of the sensitivity analysis of the attention-based explainer are depicted in Fig. 6. As illustrated, the proposed explainer successfully identifies the highly influential predictors governing the behaviour of GeoPIP. For both air temperature and wind speed prediction tasks, masking the top-ranked features consistently produced substantial degradation in estimation performance across all horizons. For air

Table 4 Component-level sensitivity analysis

Case	Air Temperature						Wind Speed					
	RMSE	MAE	R^2	WI	AbsPBIAS	EVS	RMSE	MAE	R^2	WI	AbsPBIAS	EVS
GeoPIP	1.909	1.497	0.929	0.982	2.782	0.941	0.537	0.420	0.488	0.866	3.034	0.511
1	2.188	1.810	0.792	0.949	5.420	0.877	0.515	0.403	0.486	0.861	3.654	0.533
2	6.768	5.449	-0.902	0.570	8.285	-0.515	0.877	0.678	-0.457	0.477	4.911	-0.372
3	1.813	1.748	0.857	0.963	3.767	0.900	0.527	0.415	0.455	0.855	4.697	0.552

temperature prediction, the elimination of the most influential features increased the RMSE from approximately 1.673 to 2.577 for the first prediction step and from 1.411 to 2.657 for the ninth step, while simultaneously reducing the R^2 values from 0.847 to 0.638 and from 0.943 to 0.796. Similar degradation was observed in the additional evaluation metrics, where the WI decreased from 0.964 to 0.903 and from 0.985 to 0.938, the EVS decreased from 0.874 to 0.742 and from 0.958 to 0.864, while the AbsPBIAS increased from 3.682 to 7.275 and from 2.564 to 5.301 for the first and ninth prediction steps, respectively.

Similar degradation patterns were observed across intermediate horizons, particularly at the sixth and seventh steps, where the elimination of the top-attended features caused RMSE increases exceeding +0.87 and substantial reductions in explanatory capability. For wind speed prediction, the masking operation also resulted in noticeable performance deterioration, with the RMSE increasing from 0.463 to 0.614, the R^2 decreasing from 0.625 to 0.343, the WI decreasing from 0.878 to 0.720, and the EVS decreasing from 0.630 to 0.395 for the ninth prediction step. Furthermore, the AbsPBIAS increased from 1.440 to 5.004, indicating a substantial increase in prediction bias following feature elimination.

As depicted in Fig. 6, the divergence between the GeoPIP predictions and the feature-masked predictions became increasingly pronounced across multiple horizons, indicating substantial reliance on the extracted explanatory variables. These findings demonstrate that the proposed attention-based explainer does not assign stochastic or arbitrary importance weights but instead captures physically and statistically

meaningful predictors that are intrinsically connected to the prediction mechanism of GeoPIP.

6 Conclusion

Weather prediction is crucial for early preparedness, environmental monitoring and decision-making in the event of severe and anomalous atmospheric conditions. However, predicting weather variables is challenging due to the nonlinear behaviour of atmospheric dynamics, the spatial variability of geographically distributed weather stations and the difficulty of maintaining reliable prediction performance across long prediction horizons. In this article, we proposed a geo-based physics-informed predictor for long-horizon weather estimation. The proposed system performs multi-horizon dual weather variable estimation for air temperature and wind speed by jointly exploiting geographical representations, physics-informed constraints and temporal atmospheric patterns.

The outcomes of an extensive and comparative evaluation of real measurements revealed that the introduced scheme could achieve competitive air temperature prediction and competitive wind speed prediction performance with relatively low computational overheads. In particular, GeoPIP achieved an RMSE of 1.909 and an MAE of 1.497 for air temperature prediction, and an RMSE of 0.537 and an MAE of 0.420 for wind speed prediction. Its strategic use of geo-based atmospheric representation and physics-informed learning under the proposed methodology provided superior

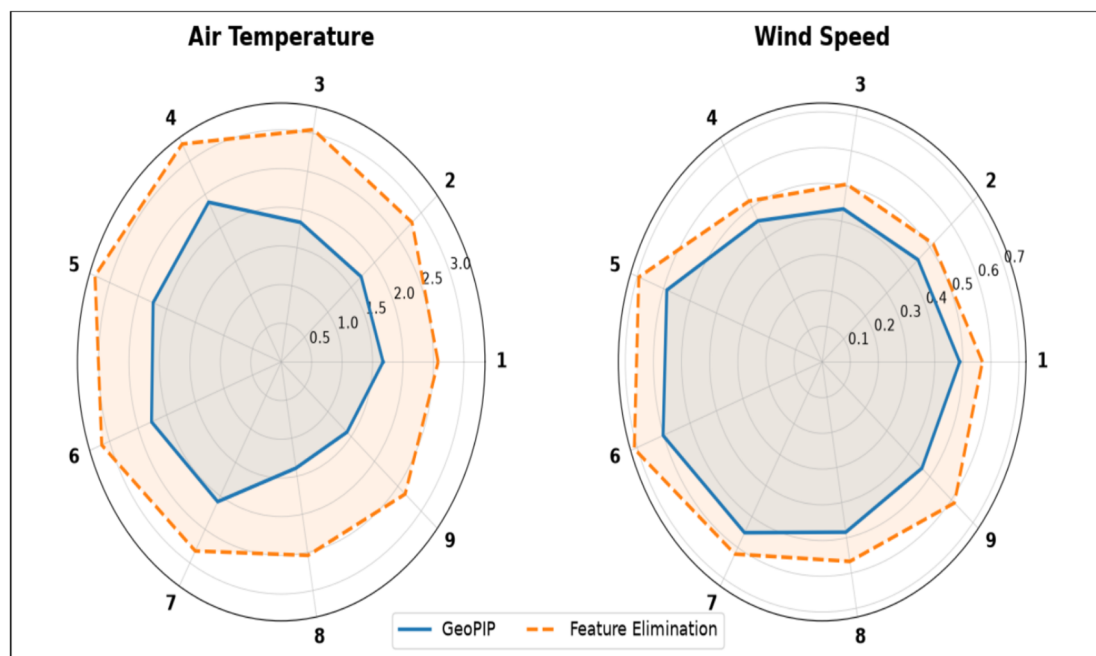


Fig. 6 Sensitivity analysis of the attention-based explainer

explainability through attention-based interpretation of the spatiotemporal prediction behaviour and the most influential atmospheric features. Thus, it can benefit the design of next-generation weather prediction systems.

Despite its promising performance, GeoPIP is subject to certain limitations. Wind speed prediction remains inherently more challenging than air temperature prediction, owing to its greater variability and stronger dependence on local atmospheric dynamics. While GeoPIP attains competitive performance in wind speed estimation, further gains could potentially be achieved through the incorporation of additional atmospheric variables and higher-resolution meteorological observations. Moreover, the application of GeoPIP to substantially larger meteorological datasets, characterised by a greater number of stations, longer historical records, and finer temporal resolution, may necessitate further optimisation to enhance computational scalability without compromising predictive accuracy.

For future work, we are currently exploring the use of quantum machine learning strategies in our proposed prediction system. Unlike traditional computers, which rely on physical 0 and 1 states, quantum computers use qubits that can simultaneously represent both $|0\rangle$ and $|1\rangle$ states. This allows for the concurrent execution of multiple computational processes (Althobaiti and Dohler 2021; Lamichhane and Rawat 2025). We anticipate that employing quantum machine learning will improve the learning efficiency of our GeoPIP, potentially enabling us to achieve the same prediction performance with fewer training samples or simpler architectures. The use of quantum machine learning techniques may also reduce computational complexity and speed up the prediction process for scalable meteorological systems.

Author Contributions A.A. conceived the study, designed the methodology, collected and preprocessed the data, developed the GeoPIP model, conducted the experiments, analysed the results, prepared the figures and tables, and wrote and revised the manuscript. The author reviewed and approved the final manuscript.

Funding This research was funded by the Taif University, Saudi Arabia. The author would like to acknowledge the Deanship of Graduate Studies and Scientific Research, Taif University, for funding this work.

Data Availability The dataset used in this study is publicly available from the King Abdullah Petroleum Studies and Research Center (KAP-SARC) Saudi Hourly Weather Data platform. The GeoPIP source code is available at: <https://github.com/Ahlam-Althobaiti/GeoPIP>.

Declarations

Competing Interests The authors declare no competing interests.

Ethics Declaration Not applicable.

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